

1. INTRODUCTION

This manual covers the RoboInject-300 injection molding machine (serial prefix SM-). Read all safety instructions before servicing. The RoboInject-300 is a fully electric injection molding machine with servo-driven injection arm and barrel heater zones. Error codes in this manual are specific to the RoboInject-300 and its control system version 4.3+. Do not apply these codes to other equipment without confirming the model.

1.1 SYMBOLS USED

WARNING ? Risk of electrical shock or personal injury.

CAUTION ? Risk of damage to equipment.

NOTE ? Additional information or tip.

1.2 MAINTENANCE SCHEDULE

Daily: check oil level in the lubrication reservoir, inspect fan intake.

Weekly: clean air filter, verify NTC sensor readings are within range.

Monthly: full calibration check, belt tension inspection.

4. ERROR CODE REFERENCE

4.1 E101 ? Winding Overtemperature

Meaning: The temperature of the injection servo motor windings has exceeded the safe operating threshold (130°C). The controller automatically reduces current to prevent thermal damage.

Probable causes:

- Blocked cooling fan intake (most common)
- Dirty or clogged air filter restricting airflow
- Failed NTC temperature sensor on the winding
- Excessive duty cycle without cooldown period

Corrective action:

Step 1. Power down the machine and lock out the main disconnect.

Step 2. Inspect the cooling fan on the back of the servo drive. Remove any debris.

Step 3. Replace the air filter if visibly dirty (P/N AF-300-12).

Step 4. Verify the NTC sensor: measure resistance between pins 3 and 4 on the sensor harness connector.

Expected value at 25°C: 2.4k $\pm$ 5%.

Step 5. If sensor is OK but problem persists, check 24V DC supply to fan motor (terminal block TB2, pins 1-2).

Step 6. Restart and run under load for 20 minutes. Monitor winding temperature on the controller display.

Source: Section 4.2, page 214. Referenced in: SS-300 Quick Reference, §E.

4.2 E204 ? Servo Stall Detected

Meaning: The injection arm servo motor has exceeded its torque limit and the controller has triggered a stall protection. The arm may be mechanically bound or the servo drive parameters are incorrect.

Probable causes:

- Mechanical obstruction in the injection arm rails
- Worn or damaged ball screw
- Servo drive current limit set too low for the mold being used
- Incorrect acceleration/deceleration ramp values

Corrective action:

Step 1. Clear the fault on the HMI (Main Menu > Faults > Reset).

Step 2. Manually jog the injection arm to full retract and full extend. Listen for grinding or binding.

Step 3. Inspect the ball screw for wear or contamination.

Step 4. Check servo drive parameter P-07 (peak current limit). Default is 12A. Increase to 14A if running heavy molds.

Step 5. Verify the acceleration ramp in parameter P-12 is not set below 500ms.

Source: Section 8.3, page 312.

4.3 E302 ? Zone 1 Heater Overcurrent

Meaning: Electrical current draw on barrel zone 1 exceeds 20A. Indicates a shorted heating band or damaged wiring.

Corrective action:

Step 1. Switch off zone 1 from the controller.

Step 2. Measure resistance across heating band terminals. Should read 11.5 ± 1 .

Step 3. Replace the heating band (P/N HB-1-300) if out of tolerance.

6. SENSOR CALIBRATION & MAINTENANCE

6.1 NTC Temperature Sensor (Winding)

The winding temperature sensor is a negative-temperature-coefficient (NTC) thermistor. Use the following table to verify its reading:

Temperature	Resistance (nominal)
25°C	2.4 k?
50°C	1.1 k?
75°C	0.55 k?
100°C	0.28 k?

If the measured resistance deviates by more than 8% from nominal, replace the sensor (P/N NTC-R300). Ensure the sensor tip is fully inserted into the thermal well and coated with heat-conductive paste.

6.2 Thermocouple (Barrel Zones)

Each barrel zone uses a Type-K thermocouple. Cold junction compensation is handled automatically by the controller board. If zone temperature readings are erratic, swap the thermocouple with an adjacent zone to isolate the fault.

6.3 Lubrication System

The central lubrication pump delivers oil to the ball screw and linear rails. Low oil level triggers alarm A107 on the HMI. Use only ISO-VG-68 grease. The reservoir holds 1.5 liters. Refill port is located behind the front access panel.

7. ELECTRICAL SYSTEM

7.1 Main Fuse Ratings

Main AC input: 3 × 32A (class gG, P/N F-M300-MAIN)

Servo drive: 2 × 16A (class aM, P/N F-M300-SERVO)

Control circuit: 1 × 6.3A (class gG, P/N F-M300-CTRL)

Heater zones: 1 × 25A per zone (class gG)

Always disconnect mains power before accessing the fuse panel.

7.2 Control Board LEDs

D1 (green) ? Power present

D2 (yellow) ? Communication active

D3 (red) ? Fault latch (see error code)

D4 (amber) ? Maintenance reminder

7.3 Safety Interlock Circuit

All access doors are monitored by magnetic safety switches (MSS-300). The interlock circuit must be closed for the machine to run. If the controller shows 'E011 ? Safety Loop Open', test each door switch by cycling it open/closed while watching the input status on the HMI (Diagnostics > I/O Map). Replace any switch that fails to toggle.